INCH-POUND
MIL-B-18/23E
W/AMENDMENT 1
25 June 2021
SUPERSEDING
MIL-B-18/23E
22 November 2010

MILITARY SPECIFICATION SHEET

BATTERY, DRY, BA-48

INACTIVE FOR NEW DESIGN AFTER
10 AUG 1998 AND IS NO LONGER USED, EXCEPT
FOR REPLACEMENT PURPOSES.

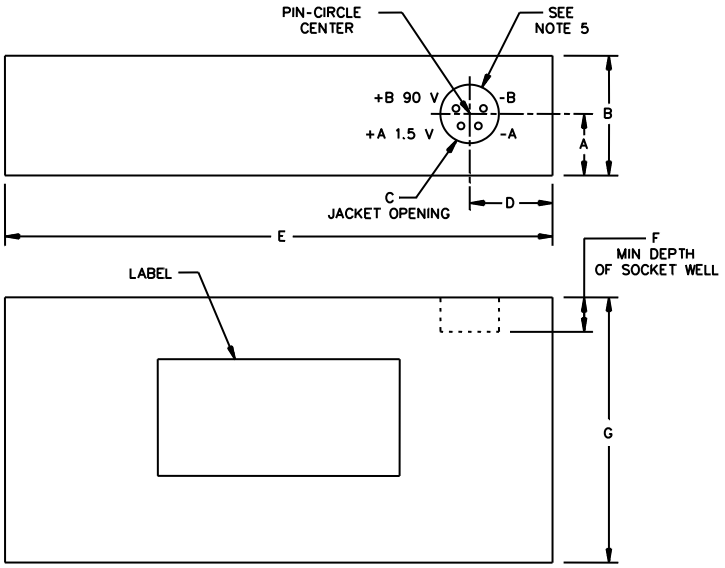
This specification is approved for use by all Departments and Agencies of the Department of Defense.

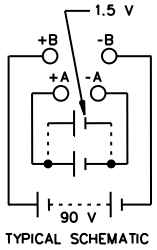
The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-B-18.

AMSC N/A

FSC 6135

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TYPICAL SCHEMATIC

| Ltr | Inches | | mm | |
|---|---|---|---|---|
| | Min | Max | Min | Max |
| A | 1.089 | 1.144 | 27.66 | 29.05 |
| B | 2.125 | 2.219 | 53.97 | 56.36 |
| C | 1.058 | 1.068 | 26.87 | 27.12 |
| D | 1.495 | 1.510 | 37.97 | 38.35 |
| E | 9.933 | 9.943 | 252.30 | 252.55 |
| F | .620 | .630 | 15.75 | 16.00 |
| G | 4.808 | 4.818 | 122.12 | 122.38 |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.005 (0.13 mm) for three place decimals and ±.02 (0.5 mm) for two place decimals.
5. Socket type X111 is required. (see MIL-B-18 for details.)
4. Marking to be in accordance with MIL-B-18.

FIGURE 1. Battery dimensions.



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REQUIREMENTS:

Dimensions, marking, and configuration: See figure 1.

| | A unit | B unit |
|---|---|---|
| Nominal voltage: | 1.5 volts | 90 volts |
| Usual number and type of cells: | 6 “CD” cells | 90 “A” cells |
| Usual cell connection: | Parallel | Series |

Terminals: Four hole socket.

Weight (maximum): 5 pounds, 8 ounces.

Capacity tests: When the battery is tested in accordance with the methods of examination and test of this specification, the minimum capacity-test requirements shall be not less than the minimum time specified for SLD or SLT.

First article inspection in accordance with MIL-B-18:

Visual and mechanical (external).
Battery voltage.
Vibration test.
Mechanical-shock test in accordance with MIL-STD-202-213, test condition I.
Insulation resistance test in accordance with MIL-STD-202-302, test condition B, tolerance ±20 volts.
Capacity, D (without storage).
Jacket integrity test.

Conformance inspection in accordance with MIL-B-18:

Visual and mechanical (external).
Battery voltage.
Insulation-resistance test in accordance with MIL-STD-202-302, test condition B, tolerance ±20 volts.
Vibration test.
Mechanical-shock test in accordance with MIL-STD-202-213, test condition I.
Jacket integrity test.
Electrolyte leakage.
Capacity, D and T.

Method of examination and test:

Capacity tests: See requirements for capacity specified herein.

Storage:

| Test | Period |
|---|---|
| D | 12 months |
| T | 90 days |

Discharge: The “A” unit shall be discharged through 5 ohms and the “B” unit through 9,000 ohms for 5 hours per day, 5 days per week. This cycle shall be repeated continuously to a test-end voltage of 1.1 volts for the “A” unit and 65.0 volts for the “B” unit.



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Closed circuit voltage: Use TS-183 ( )/U test per special marking or, with minimum permissible voltage as specified for TS-183 ( )/U, use a load resistance value as noted below:

| | |
|---|---|
| A section | 1.998 ohms |
| B section | 3,440 ohms |

Special marking on each unit package:

TO TEST THIS BATTERY WITH TS-183 ( )/U:

| USE JACK NO. | MINIMUM PERMISSIBLE VOLTAGE |
|---|---|
| Unit A – 1 | 1.35 |
| Unit B - 26 | 84.0 |

Unless expressly authorized, this test information shall apply to Army and Air Force applications only.

Part or Identifying Number (PIN): BA-48, see 1.2 of MIL-B-18.

Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Referenced documents. In addition to MIL-B-18, this document references the following:

MIL-STD-202-213 MIL-STD-202-302

Custodians:
Army - CR
Navy - SH
DLA - CC

Preparing activity:
DLA - CC

(Project 6135-2021-003)

Review activities:
Army - CR4
Navy - MC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.



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